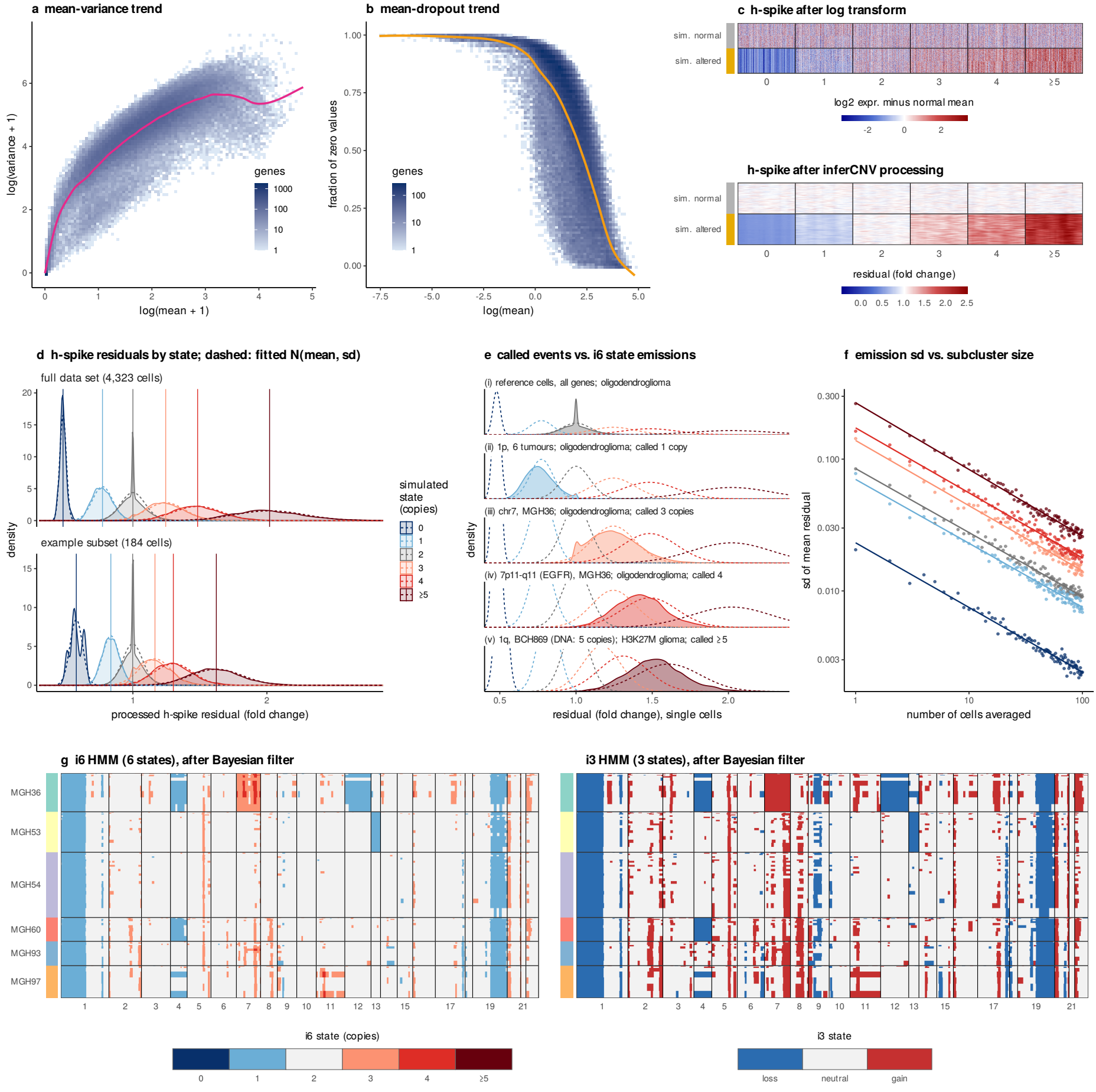

a mean-variance trend
log(variance + 1)
log(mean + 1)
genes
1000
100
10
1
b mean-dropout trend
fraction of zero values
log(mean)
genes
100
10
1
c h-spike after log transform
sim. normal
sim. altered
log2 expr. minus normal mean
h-spike after inferCNV processing
residual (fold change)
d h-spike residuals by state; dashed: fitted N(mean, sd)
full data set (4,323 cells)
example subset (184 cells)
density
processed h-spike residual (fold change)
simulated state (copies)
e called events vs. i6 state emissions
(i) reference cells, all genes; oligodendroglioma
(ii) 1p, 6 tumours; oligodendroglioma; called 1 copy
(iii) chr7, MGH36; oligodendroglioma; called 3 copies
(iv) 7p11-q11 (EGFR), MGH36; oligodendroglioma; called 4
(v) 1q, BCH869 (DNA: 5 copies); H3K27M glioma; called ≥5
residual (fold change), single cells
f emission sd vs. subcluster size
sd of mean residual
number of cells averaged
g i6 HMM (6 states), after Bayesian filter
MGH36
MGH53
MGH54
MGH60
MGH93
MGH97
i6 state (copies)
i3 HMM (3 states), after Bayesian filter
i3 state
loss
neutral
gain